

Resistivity Anisotropy and Resistivity under Biaxial Strain in CeRh_2As_2

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Resistivity Anisotropy — Motivation

CeRh_2As_2 is an actively studied unconventional superconductor. Despite a steady improvement in the quality of available crystals, the residual resistivity remains at a high value of $>30 \mu\Omega\text{cm}$. Finding the dominant types of crystallographic defects may help with further improvements of crystal quality.

- Key observation:** the residual resistivity ratio (RRR) is here defined as:

$$RRR = \rho_{300\text{ K}} / \rho_{0.5\text{ K}}$$

and can be used as a measure of crystal quality. RRR correlates with the critical temperature of superconductivity (T_c) and a magnetic order (T_0).

Crystallographic data indicates a larger uncertainty for the c -axis lattice constant [1].

- Possible explanation:** stacking faults between closely related CaBe_2Ge_2 - and ThCr_2Si_2 -type structures could increase residual resistivity along the c -axis due to additional disorder.

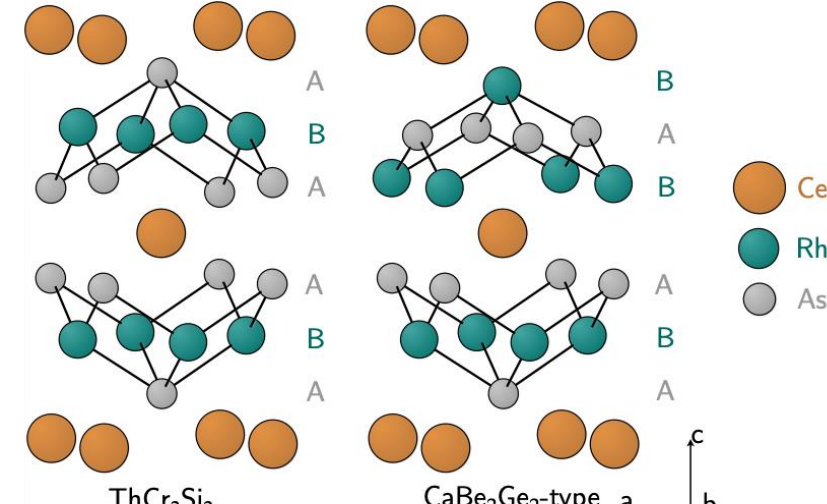


Fig. 1: ThCr_2Si_2 - and CaBe_2Ge_2 -type structures. CeRh_2As_2 crystallizes in the CaBe_2Ge_2 -type structure. After Fig. 1(a) in Ref. [2].

- Aim:** measure resistivity anisotropy to test this hypothesis (by comparing how RRR_{ab} and RRR_c change with crystal quality).

Method

Initial characterization: RRR values of pristine crystals from different batches were obtained.

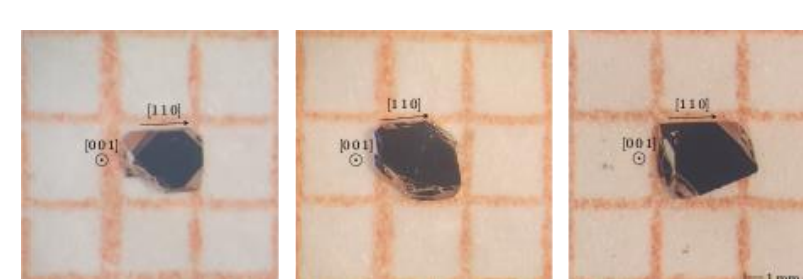


Fig. 2: Single crystals of CeRh_2As_2 . From left to right: sample 2, 1 & 3

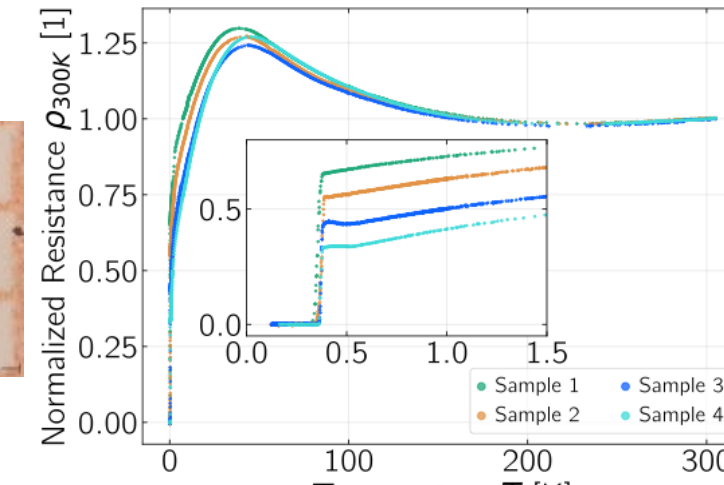


Fig. 3: Reference measurement, normalized to 300 K.

- Microstructuring:** lamellae are made using Xe-plasma focused ion-beam (FIB) milling. They are then placed on sapphire substrates, anchored by Pt deposition and Au/Ti layers sputtered on top. The Au/Ti layers are used for defining electrodes.

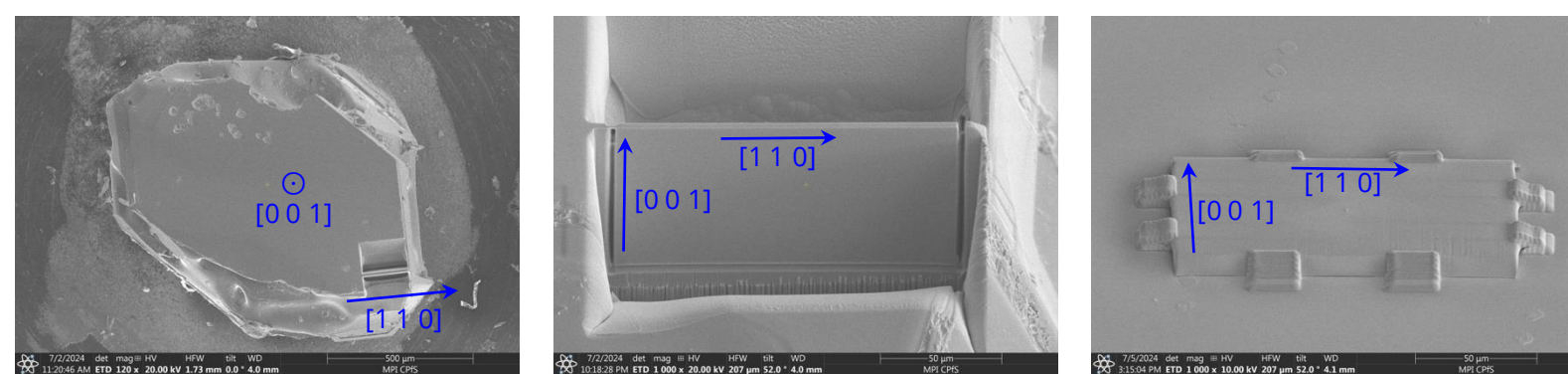


Fig. 4: Stages of Xe-plasma FIB milling of Lamella.

- Device:** inside each lamella, a microstructure is milled to make a device for c - and ab -axis resistivity measurements. The design includes meandering paths for strain relief.

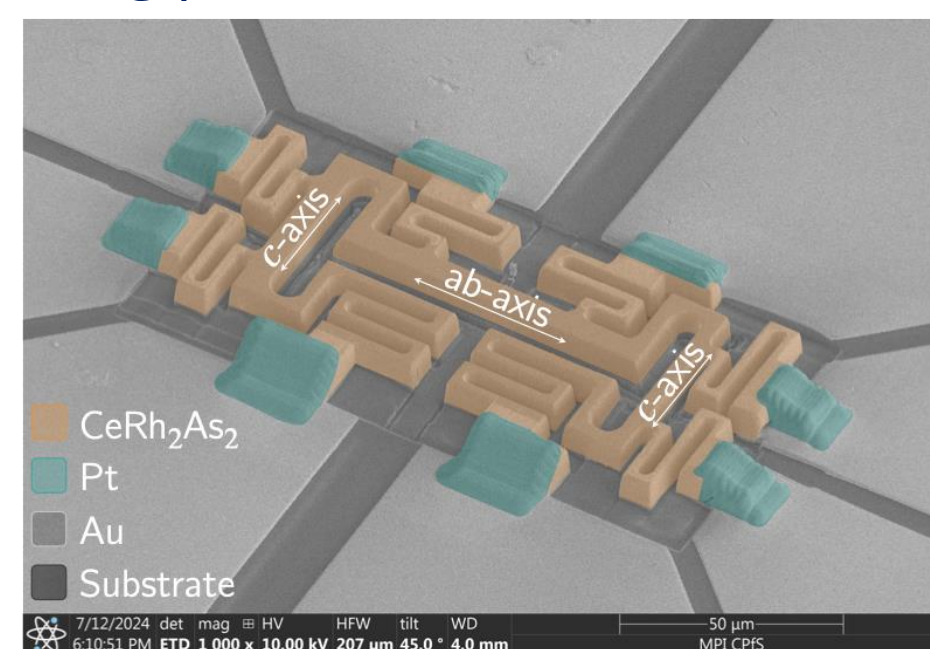


Fig. 5: Microstructured resistivity-anisotropy device from sample 1.

Key Results & Conclusions

- ab -plane resistivity is comparable to the reference measurement. RRR does not change after FIB exposure.

Batch ID	RRR ($\rho_{300\text{ K}} / \rho_{0.5\text{ K}}$)			
	Bulk	ab -axis	c -axis (thin)	c -axis (wide)
Sample 1	1.504	1.441	1.176	1.189
Sample 2	1.793	1.832	1.493	1.507
Sample 3	2.242	2.475	1.856	1.898
Sample 4	2.990			
Mishra, et al. [3]		2.351	1.543	

- Superconductivity and T_0 suppressed.

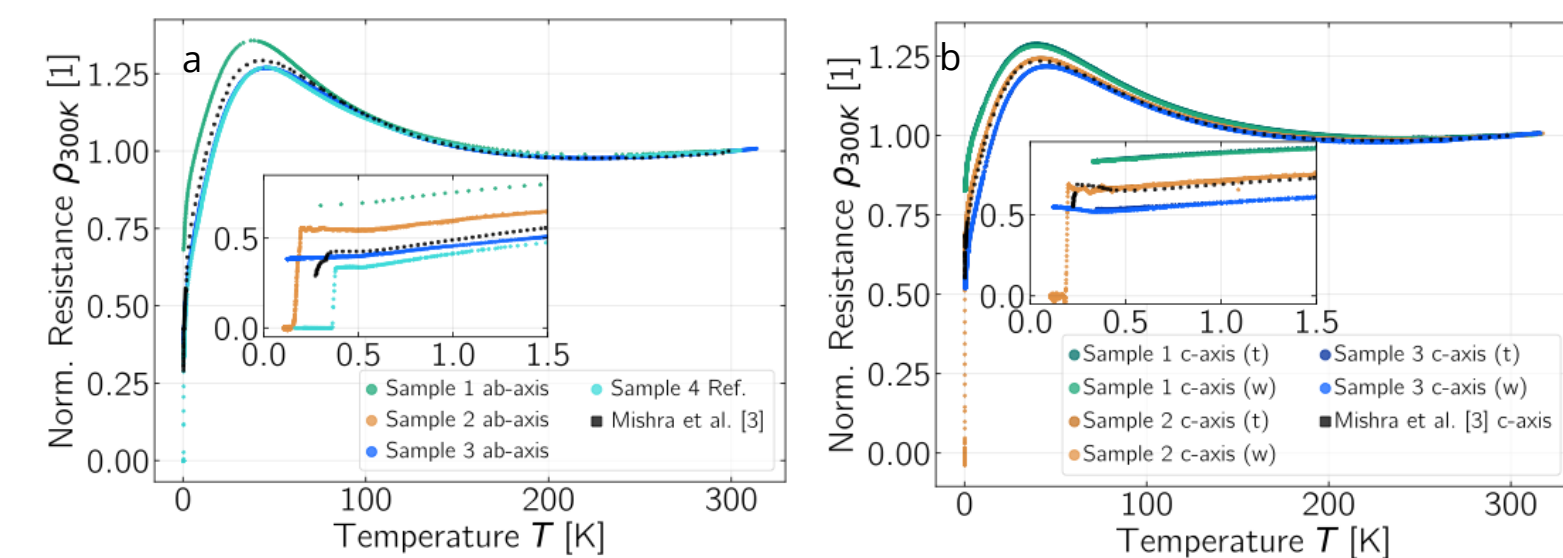


Fig. 6: in-plane and c -axis resistivity normalized to 300 K (a) a -axis & ab -axis. (b) c -axis.

- RRR_c and RRR_{ab} improve with crystal quality at a comparable rate. The suppression of T_c suggests that the effects of FIB irradiation or substrate-induced strain may be non-negligible.

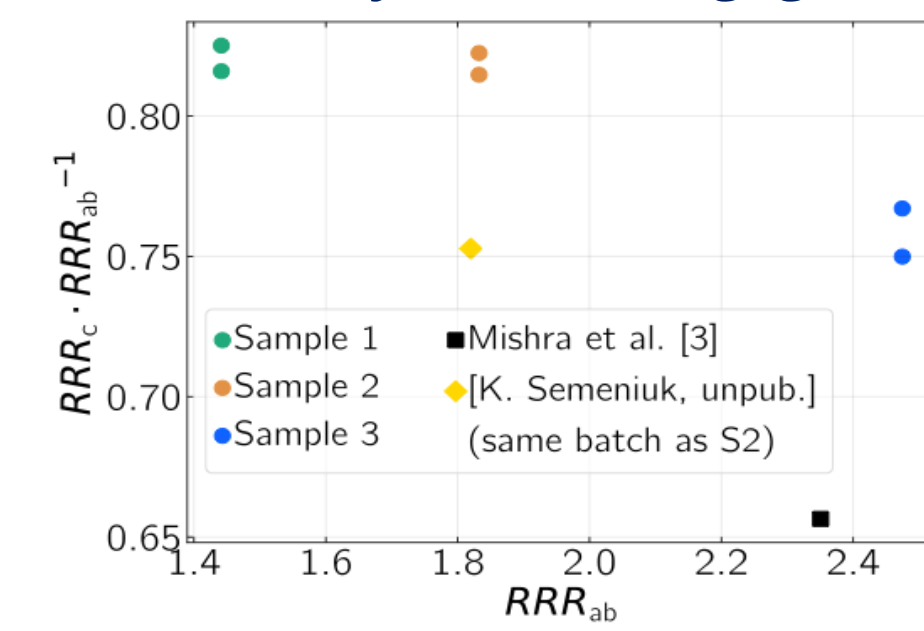


Fig. 7: Resistivity anisotropy as RRR . The ratio RRR_c / RRR_{ab} against RRR_{ab} .

Biaxial Strain — Motivation

Many strongly correlated electron systems can be effectively tuned via pressure or strain, but conventional strain cells are often impractical for sub-kelvin experiments due to size and thermalization issues. Substrate-induced strain offers a practical alternative.

- Idea:** apply biaxial strain by gluing a polished crystal to a substrate, using differential thermal expansion during cooldown.
- Substrate choice:** silicon for low thermal contraction and high stiffness ($E \approx 172 \text{ GPa}$); estimated strain limit $\epsilon_{max} \approx 0.28\%$.
- COMSOL FEM:** simulated setup to estimate biaxial strain magnitude and distribution.
 - Predicted in-plane strain up to $\epsilon_{max} \approx 0.29\%$, with a 15% variation near the probes and 3% through the crystal thickness at the center.

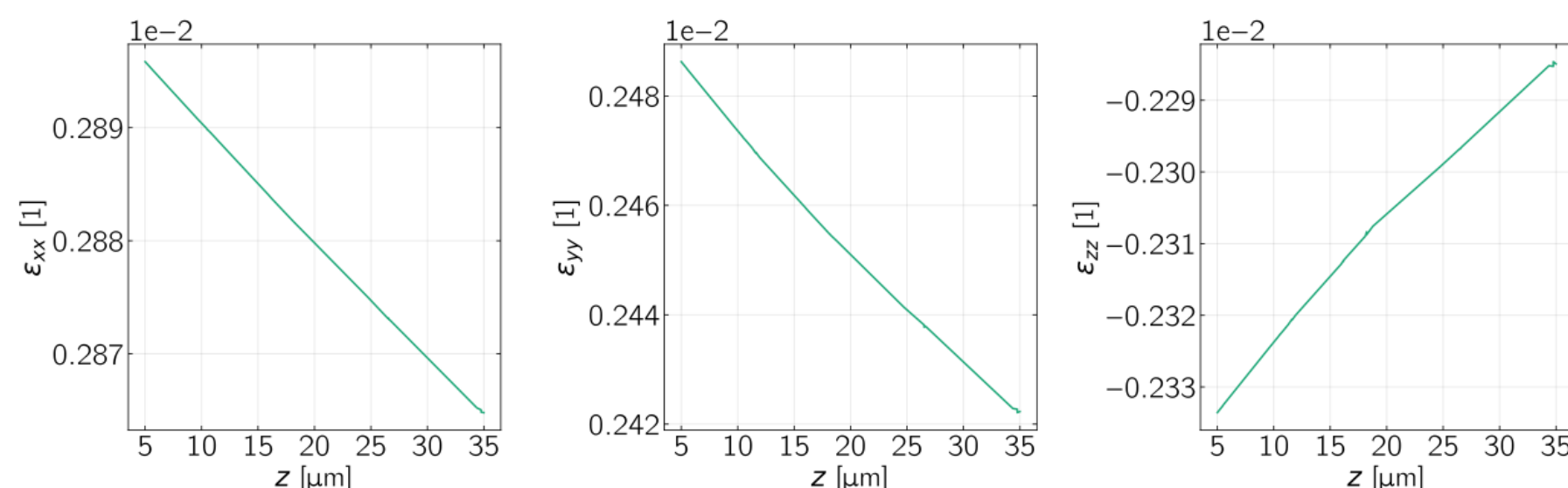


Fig. 8: COMSOL simulation of strain components through the crystal thickness at the center of the crystal (z -direction). Geometry based on a crystal from batch 3.

- A high aspect ratio between the lateral size and thickness is required for homogeneous biaxial strain.

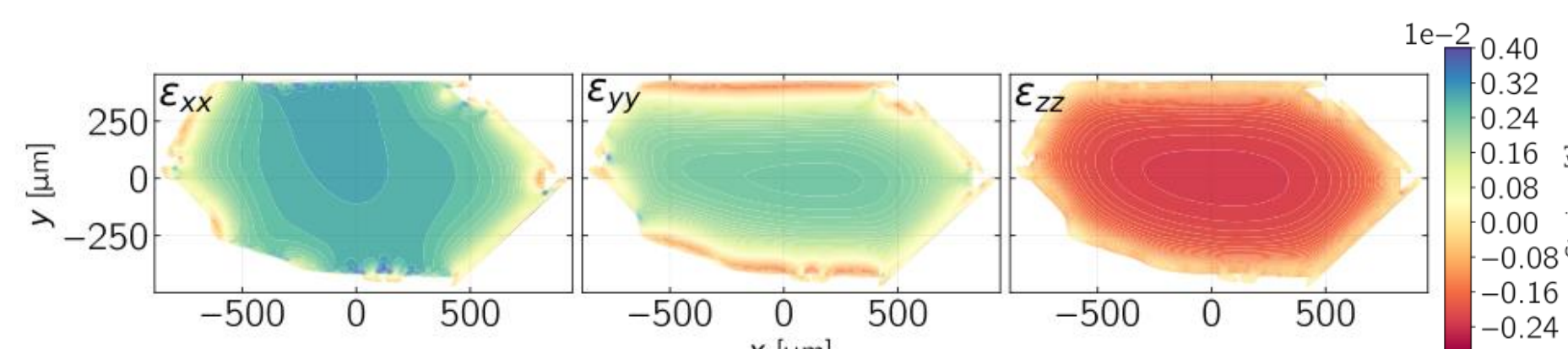


Fig. 9: COMSOL simulation of the strain components on the top surface of the crystal. Geometry based on a crystal from batch 3

Uncertainties & limitations: elastic constants of the crystal are placeholders ($E_{a,b} = 129 \text{ GPa}$, $E_c = 109 \text{ GPa}$) taken from [4], elastic constants of adhesive are taken from Stycast 2850FT ($E = 6.413 \text{ GPa}$).

Method

- Crystals are polished using a hand-polishing jig.
- Optical profilometry is used to determine crystal height. Final thickness $\approx 30 \mu\text{m}$.

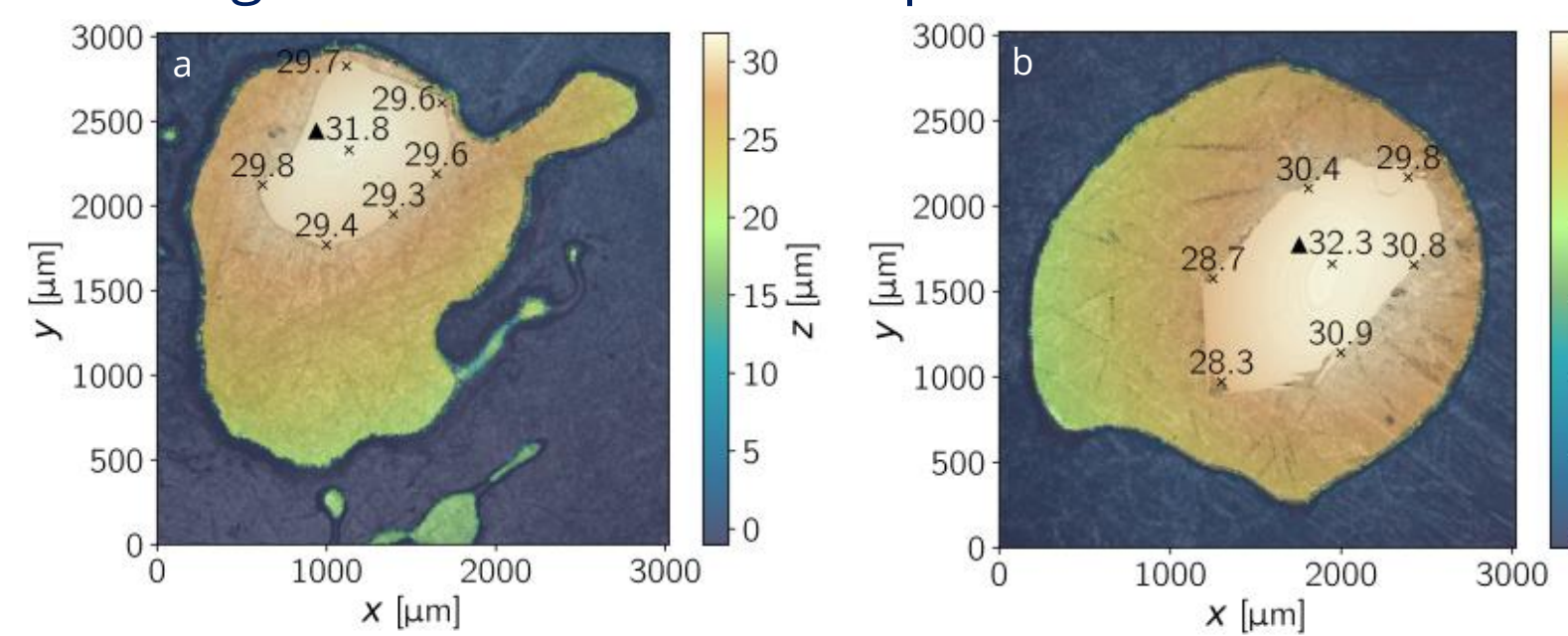


Fig. 10: Height maps from optical profilometry of polished crystals. (a) Batch 4. (b) Batch 3.

- The polished crystal is mounted to a substrate with Stycast 1266 epoxy.
- After curing, the crystal is cleaned and re-measured, giving an estimated adhesive thickness of 3–5 μm .

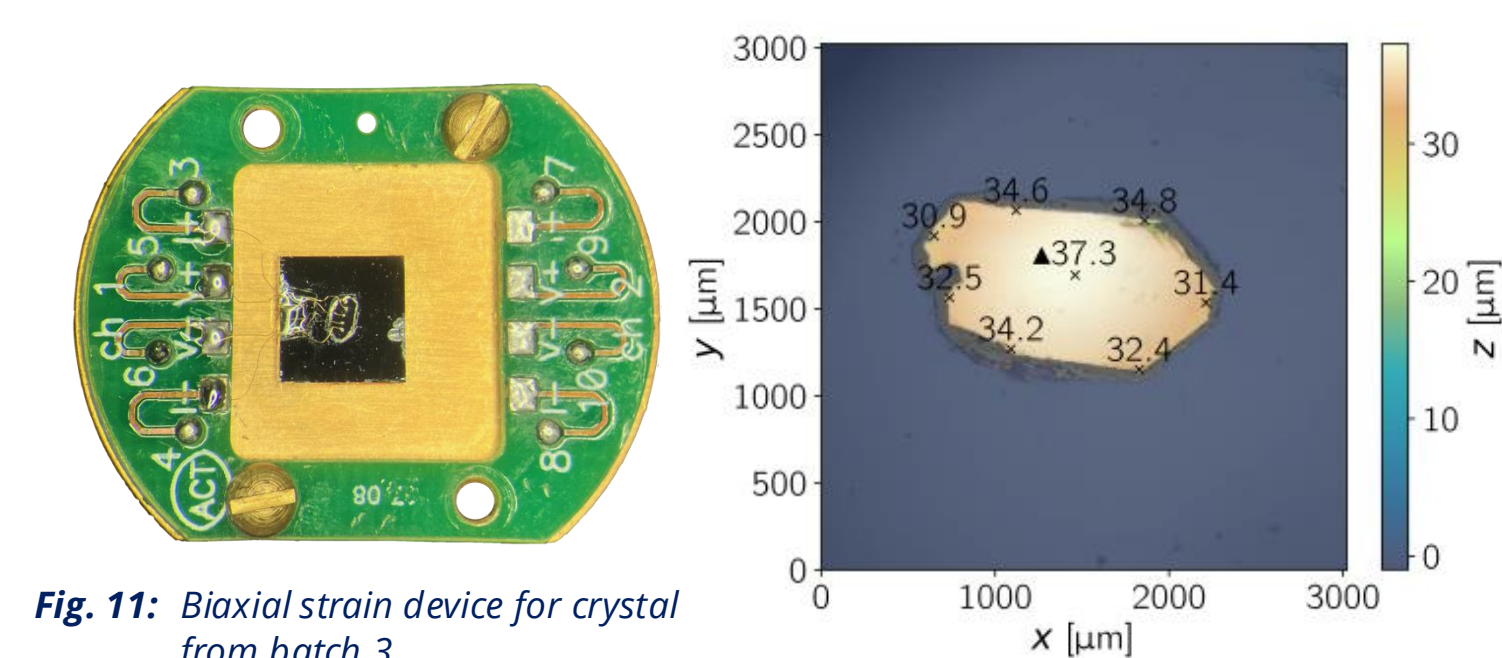


Fig. 11: Biaxial strain device for crystal from batch 3.

Fig. 12: Height map from optical profilometry for mounted crystal from batch 3.

Preliminary Results

- Measurement:** superconductivity and T_0 suppressed; for $T > 0.75 \text{ K}$ there is no change in the temperature dependence of resistivity (compared to ab -axis).

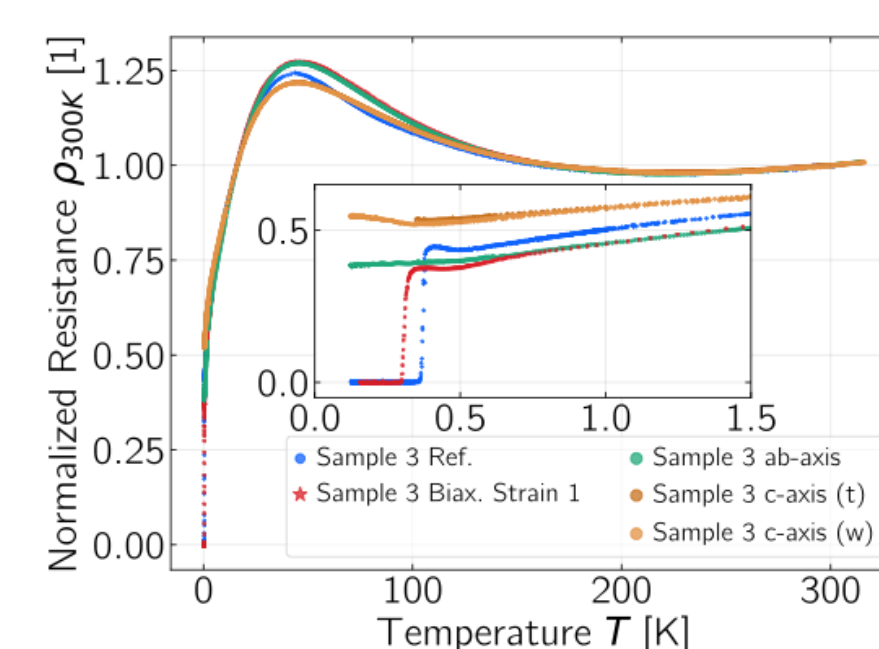


Fig. 13: Resistivity from biaxial strain measurement (batch 3) vs. reference and anisotropy data for same-batch crystals, normalized to 300 K.

- Conclusion:** established a feasible sample preparation protocol for applying substrate-induced biaxial strain. The observed effects were likely reduced due to poor strain transmission through the adhesive.

References:

- [1] P. Khanenko et al., "Phase diagram of CeRh_2As_2 for out-of-plane magnetic field," *Phys. Rev. B*, vol. 112, L060501, Aug. 2025.
- [2] S. Khim et al., "Field-induced transition within the superconducting state of CeRh_2As_2 ," *Science* 373, 1012-1016 (2021).
- [3] S. Mishra et al., "Anisotropic magnetotransport properties of the heavy-fermion superconductor CeRh_2As_2 ," *Phys. Rev. B* 106, L140502 (2022).
- [4] Maja D. Bachmann et al., "Spatial control of heavy-fermion superconductivity in CeIrIn_5 ," *Science* 366, 221-226 (2019).

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